Efficient Hardware Architecture for SHA3 Hash Function Implementation in CRYSTALS-Kyber Encryption

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***Abstract*—As quantum computing advances rapidly, post-quantum cryptography (PQC) has gained significant attention. Among the proposed schemes, the lattice-based CRYSTALS-KYBER algorithm has been selected by NIST in 2022 as the standard key encapsulation mechanism scheme. In CRYSTALS-KYBER, the SHA-3 hash function, particularly the Keccak permutation core, dominates both computational complexity and execution time. To improve efficiency, this paper proposes a SHA-3 hardware architecture customized for CRYSTALS-KYBER, supporting multiple security levels and integrating a rejection sampling mechanism to meet the functional requirements of the Scalable Output Function (XOF). Implemented in Verilog RTL and synthesized using a TSMC 40 nm process technology, the proposed architecture achieves a frequency of 400 MHz. The experimental results demonstrate that the design achieves an optimal balance between area and performance, with a reduced gate equivalent (GE) count compared to existing works.**

Keywords—Digital circuits, Hardware architecture, Hash function, Post-quantum cryptography.

I. Introduction

In 2012, the National Institute of Standards and Technology (NIST) began a public evaluation to standardize post- quantum public-key encryption algorithms. Lattice-based algorithms gradually became the leading candidates due to their strong security and performance. In 2022, NIST selected CRYSTALS-KYBER as the first standardized post-quantum key encapsulation mechanism scheme.

Reference [1] describes the architecture of CRYSTALS-KYBER as part of the third round submission to the NIST standardization process. CRYSTALS-KYBER supports three security levels: Kyber-512, Kyber-768, and Kyber-1024. The primary difference among these levels is the choice of the parameter k, which directly affects the number of required polynomial vectors. These security levels are based on the polynomial ring $Rq = Zq\,[X]\,/\,(X^n + 1)$, where in CRYSTALS- KYBER, q = 3329 and n = 256. The functions PRF, XOF, H, G, and KDF used in CRYSTALS-KYBER are instantiated according to the FIPS-202 [2] SHA-3 standard. In the CRYSTALS-KYBER scheme, Keccak is the most time-consuming part due to its use of the SHA-3 hash function. To improve performance, a dedicated SHA-3 hardware design has been developed for use in the CRYSTALS-KYBER system.

The primary objective of this work is to design a SHA-3 hardware circuit tailored specifically to the requirements of CRYSTALS-KYBER, aiming to enhance the computational efficiency in hardware implementations and address the performance bottleneck caused by the hash operations. The contributions of this work are as follows:

1. Investigated the impact of different values of k in CRYSTALS-KYBER on SHA-3 primitives and implemented support for Kyber-512, Kyber-768, and Kyber-1024, corresponding to different security levels.
2. Introduced a rejection sampling technique compliant with CRYSTALS-KYBER (with q = 3329), improving the functionality of the SHA-3 primitives for CRYSTALS- KYBER.
3. Adopted a more compact 12-round design in the Keccak implementation to enhance circuit performance.
4. Optimized the circuit area by reducing the round constant size from 64-bits to 7-bits.

This paper aims to enhance the hardware performance of CRYSTALS-KYBER by improving the efficiency of SHA-3, which constitutes a critical component of the system in terms of execution time and resource utilization.

The remainder of this paper is organized as follows. In Section II, we revisit previous works. Section III presents the fundamental structure of SHA-3. Section IV demonstrates the proposed approach. Section V presents the experimental results. Section VI provides some concluding remarks.

II. Survey On Existing Work

Over the years, numerous studies have focused on the hardware design of the SHA-3 algorithm, aiming to enhance its throughput and reduce hardware area. For example, in [3], researchers proposed a high-throughput and low-area SHA-3 design. In addition, related works such as [4] and [5] concentrate on improving the processing speed of SHA-3 through high-performance pipeline architectures. Studies [6] and [7] implemented circuit design architectures for SHA3-512. Reference [8] explored hardware acceleration of SHA-3 on FPGA platforms and proposed several methods to enhance the computational efficiency of SHA3-512 in multimodal biometric authentication applications, including a Synchronized Padder Block and a pipelined Dual-f architecture to improve SHA-3 performance on FPGA.

However, the aforementioned studies have not focused on optimizing the hardware performance for CRYSTALS-KYBER, nor do most SHA-3 circuit designs simultaneously support multiple operation modes. Their primary objective remains the development of standalone SHA-3 implementations aimed at achieving faster computation and higher efficiency.

Reference [9] presents a SHA-3 hardware architecture designed for Kyber-768, where each module employs a finite state machine with datapath (FSMD) structure. The required mode information is stored in separate ROMs, and the control signals are managed by a dedicated controller. Implemented using Synopsys 65 nm technology, the design achieved a frequency of 312.5 MHz and an area of 340,048 μm^2. However, this architecture lacks flexibility in its padding mechanism, making it difficult to adapt to varying input lengths.

Similarly, [10] also presented a SHA-3 hardware design for Kyber-768, employing a divider-based architecture to allow the padding mechanism to dynamically accommodate different input lengths. This design, implemented using Synopsys 65 nm technology, achieved a frequency of 400 MHz, an area of 130,069 μm^2, and 67.7 kGE (where GE represents the size of a standard NAND gate).

However, neither of these works includes an architecture for rejection sampling. In the cryptographic operations of CRYSTALS-KYBER, the sampling mechanism is crucial for ensuring both security and computational efficiency. Especially in the computation of the Module-LWE (Learning With Errors) problem, the quality of random sampling directly impacts the system's security and resistance to attacks. Therefore, our work further integrates Rejection Sampling into the SHA-3 hardware architecture to enhance the overall performance of CRYSTALS-KYBER in hardware implementations.

III. Fundamental Structure

SHA-3 is a secure hash algorithm based on the Keccak algorithm. Keccak utilizes a sponge function. Fig. 1 presents the structure of the sponge function, which constitutes the core computational framework in SHA-3. After the input message is processed with the appropriate suffix and padding, it undergoes a bitwise XOR operation with the r-bit input blocks. The result is then concatenated with the c-bit and passed through the permutation function f. The output is subsequently truncated to obtain the desired number of bits. If the required output length is not met, the permutation function f is reapplied, and the output is truncated again. This process is repeated until the output length satisfies the specified requirement.

In the CRYSTALS-KYBER encryption process, rejection sampling is a critical mechanism for generating random values that fall within a valid range, typically less than the prime modulus q. Values outside this range are discarded and regenerated. This technique ensures uniform distribution and plays a vital role in maintaining the security of lattice-based cryptographic systems.

IV. The Proposed Design

This section introduces the SHA-3 scheme proposed in this paper, which is specifically designed and constructed for CRYSTALS-KYBER, as shown in TABLE I. "FUNC" represents the functions in CRYSTALS-KYBER that utilize SHA-3, while "MODE" corresponds to the specific SHA-3 algorithm used.

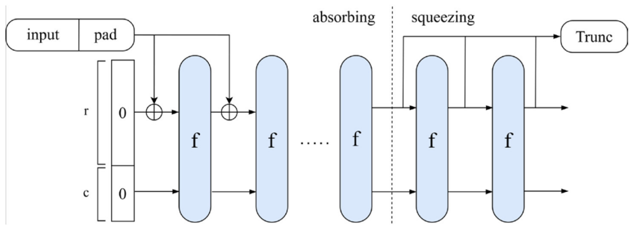

Fig. 1. Sponge function

TABLE I
Configuration mapping of SHA-3 modes and functions

SHA-3	Function	MODE	FUNC	r	r/64
SHA3 256	H(pk) H(c) H(m) G(d)	01	–	1088	17
SHA3 512	G(m||H(pk)) G(m||h)	00	–	576	9
SHAKE-128	XOF	10	–	1344	21
SHAKE- 256	PRF KDF	11	00 01	1088	17

The proposed circuit architecture is shown in Fig. 2, which consists of two main modules: Data Process and SHA-3 Permutation. TABLE II describes the external input and output ports of the proposed hardware design. The Data Process module is responsible for handling the input string. The input string is provided in 64-bit segments, which are collected and padded by the Data Process module before being passed to the SHA-3 Permutation module. The SHA-3 Permutation module performs the Keccak computation along with Rejection Sampling on the padded data.

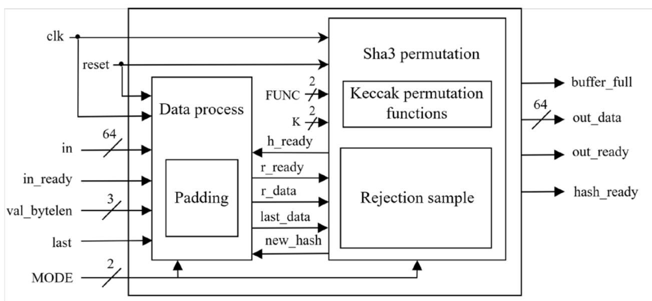

Fig. 2. Proposed SHA-3 hardware core

A. Padding

The padding logic fills input messages that do not reach the full r-bit length to ensure alignment with the SHA-3 specification. The proposed architecture supports dynamic padding by appending 64-bit segments per cycle until a complete r-bit block is formed. As shown in Table III, the `val_bytelen` signal indicates the number of valid bytes in the final input data. Fig. 3 illustrates the padding mechanism

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used in SHA-3, which consists of eight multiplexers, each responsible for processing one byte.

TABLE II
EXTERNAL I/O PORT DESCRIPTION OF THE PROPOSED SHA-3 CORE

Port	Width	Description
clk	1	Clock
reset	1	Asynchronous reset
in	64	Input data
in ready	1	Input valid
val bytelen	3	Number of valid bytes in the last input data
last	1	Last input
MODE	2	SHA-3 mode
FUNC	2	Function mode
K	2	CRYSTALS-KYBER k
buffer full	1	Buffer is full or not
out data	64	Out data
out ready	1	Output valid
hash ready	1	When set to 1, the system is ready to receive data

Based on the selected `pad_mode`, each multiplexer selects the proper padding value and uses the control signals `pad_head` and `pad_byte` to identify whether the byte corresponds to the start of the padding or a zero-filled segment. A logic low indicates the required padding header specified by SHA-3 (such as 0x06 or 0x1F), while one indicates an 8-bit zero. Additionally, a final padding bit with the value one is appended at the end of the message to comply with the hash or SHAKE block termination rule.

TABLE III
PADDING MODE CONFIGURATION

val bytelen	padmode
0	8'b11111110
1	8'b11111101
2	8'b11111011
3	8'b11110111
4	8'b11101111
5	8'b11011111
6	8'b10111111
7	8'b01111111

B. Keccak permutation functions

Keccak is the core computation unit of the SHA-3 algorithm. It corresponds to the function f in the sponge function shown in Fig. 1 and operates on a 1600-bit state. This state is composed of an r-bit message portion and a c-bit capacity portion. In SHA-3, the Keccak-*f* function performs permutations through five sequential steps: θ, ρ, π, χ, and ι, repeated over 24 rounds to ensure mixing, diffusion, and nonlinearity.

To improve efficiency, the proposed design supports a 12-round configuration as well. Fig. 4 illustrates the dataflow for one round of this configuration. A multiplexer (MUX) selects the input source: either new external data when the `sel` signal is high, or feedback from the previous round when the `sel` signal is low. This design allows the same computation unit to be reused across multiple rounds, thereby improving resource utilization.

To optimize storage and logic complexity in the Iota step of SHA-3, the 64-bit round constants (RCs) are compressed into 7-bit representations by leveraging the observation that only bit positions 0, 1, 3, 7, 15, 31, and 63 are ever set. This approach reduces memory usage and minimizes XOR operations without affecting functional correctness.

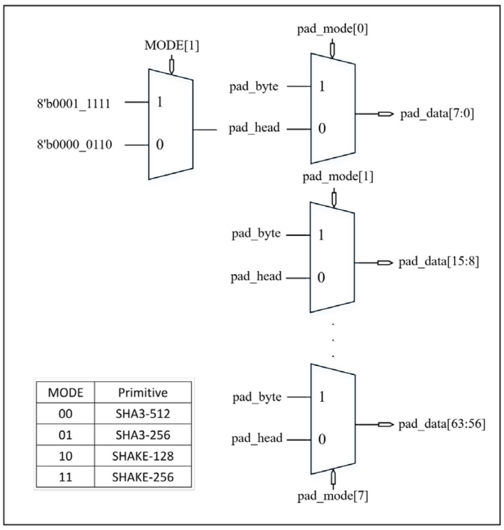

Fig. 3. The padding mechanism

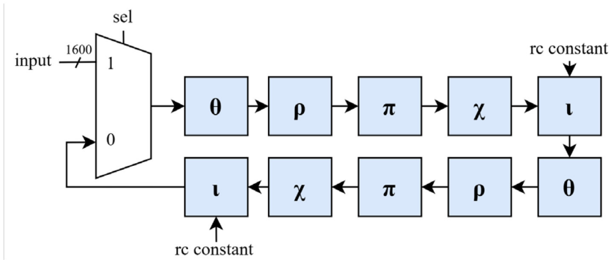

Fig. 4. Keccak structure

C. Rejection Sampling

In the CRYSTALS-KYBER encryption scheme, rejection sampling is employed during the generation of matrix A, whose elements are uniformly distributed random coefficients over a finite field. Since the CRYSTALS-KYBER specification requires that each coefficient be a 12-bit integer strictly less than where q = 3329, the design extracts 12-bit segments from the SHAKE128 output and evaluates them through the lower3329 module. Valid values are retained, while those that do not satisfy the condition are discarded, and the system proceeds to evaluate the next candidate.

In conventional designs, if the result obtained from the hash operation does not fall within the valid range less than the prime q, the central processor must issue additional computation requests to the IP and re-execute the hash function until the returned result satisfies the algorithmic requirements. This repeated interaction leads to significant overhead due to frequent interruptions and data transfers. In our proposed design, such overhead is completely eliminated by integrating the rejection sampling mechanism directly into the SHA-3 hardware architecture, thereby enhancing the overall performance in hardware implementations.

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V. Experimental Results And Comparison

In this section, we compare the implementation results of the proposed design with related works reported in the open literature. The proposed architecture is implemented in Verilog RTL and synthesized using Synopsys Design Compiler with TSMC 40nm process technology. Due to differences in technology nodes, design scopes, and optimization objectives among studies, achieving a fully fair and representative comparison is inherently challenging. To provide a meaningful and informative comparison, we compare the implemented components of the proposed design with available ASIC synthesis results from the literature. The comparison covers key metrics such as the rejection sampling unit, the Keccak-f [1600] core, and the overall SHA-3 core architecture.

As shown in Table IV, the comparison of the proposed Rejection Sampling module with those reported in previous works demonstrates that our design achieves a smaller gate equivalent (GE) count than those presented in [11] and [12]. Although the area reported in [13] is smaller than ours, it should be noted that the implementation in [13] focuses solely on the comparison logic for values less than q and does not incorporate the signal control logic related to the Keccak core in SHA-3.

TABLE IV
Comparison of Rejection Sampling Unit Implementations

Rejection Sampling (q=3329)			
References	**Tech(nm)**	**Freq(MHz)**	**Area(GEs)**
[11]*	28	300	325K
[12]	65	200	13K
[13]	65	50	4.19K
This work	40	400	10.4K

* : Total cell area of the sample unit

In the Keccak-f implementation results shown in Table V, our design achieves the best area and frequency performance among the listed works. This is primarily due to the adoption of a 7-bit round constant representation, which effectively reduces hardware resource usage and leads to area optimization.

TABLE V
Comparison of Keccak-f[1600] Hardware Implementations

Keccak-f [1600]				
References	**Tech(nm)**	**Freq(MHz)**	**Area(GEs)**	**Round**
[11]	28	300	26K	24
[12]	65	200	24K	24
[14]	40	72	34.5K	24
This work	40	400	22.5K	24
This work	40	400	47K	12

The ASIC implementation results based on the TSMC 40nm process are summarized in Table VI. Compared to [9] and [10], our design achieves a smaller gate equivalent (GE) count.

Moreover, while the implementations in [9] and [10] only support the Kyber-768 security level (parameter k=3), the proposed design in this work is applicable to Kyber-512, Kyber-768, and Kyber-1024. In addition, as shown in the Table VII, we also implemented versions that include rejection sampling and reduced the number of Keccak-f rounds reduced from 24 to 12. Although these configurations result in area increases of 28.53% and 65.6%, respectively, they significantly enhance the completeness of the CRYSTALS-KYBER flow and improve the overall computation time.

TABLE VI
Hardware Comparison of SHA-3 Core Designs

SHA-3 core			
References	**[12]**	**[10]**	**This work**
Tech(nm)	65	65	40
Freq(MHz)	312.5	400	400
Area(μm^2)	340048	130069	59167
Area(GEs)	–	67.7K	65.2K
Round	24	24	24
Rejection Sampling	X	X	X
k	3	3	2,3,4

TABLE VII
Comparison of SHA-3 Core Variants under Different Configurations

SHA-3 core				
Tech(nm)	40	40	40	40
Freq(MHz)	500	400	400	400
Area(μm^2)	60396	59167	76080	98420
Area(GEs)	66.5K	65.2K	83.8K	108K
Round	24	24	24	12
Rejection Sampling	X	X	✓	✓
k	2,3,4	2,3,4	2,3,4	2,3,4

VI. Conclusion

This paper presents a high-performance SHA-3 hardware core that supports SHA-3 primitives as well as Kyber-512, Kyber-768, and Kyber-1024, with integrated rejection sampling functionality. In future work, we will implement the full CRYSTALS-KYBER algorithm and further optimize the SHA-3 primitives to meet growing security and efficiency requirements in applications such as IoT.

Acknowledgment

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